

Ian Schillebeeckx

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EDUCATION

Washington University in St. Louis

PhD, Computer Science (Machine Learning) – Fastest graduation in department history

St. Louis, MO

2012 – 2016

Saint Louis University

Honors BS, Computer Science

Honors BS, Applied Mathematics

St. Louis, MO

2007 – 2011

EXPERIENCE

Director of Enterprise Data Science and AI

CareDx, Inc. (Diagnostics, ~750 employees)

January 2025 – Present

San Francisco, CA

- Built company's first centralized data function from zero; strategically owned data-driven optimization ecosystem spanning analytics, experimentation, and AI; hired, developed, and retained high-performing team of analysts and data scientists; served as central point of truth for performance across customer and operational journeys
- Led executive-level metric reviews as quantitative thought leader to CEO, CFO, and COO: translated complex data science findings into actionable business strategies; presented board-level decisions backed by rigorous modeling on growth, conversion, and revenue
- Owned conversion and revenue forecasting: custom time series optimization (ARIMA, Prophet, custom objectives) across 3/6/12/24-month horizons covering ~75% of revenue, reducing error from 6% to 0.9%; Databricks ETL pipelines; weekly C-suite reports, CFO planning, Wall Street reporting
- Drove causal analysis on full conversion and engagement funnel: applied causal DAGs, causal forests, DiD, and propensity matching to quantify marginal impact of sales and billing processes; translated findings into data-driven business rules driving 10% revenue increase (~\$25M) from conversion rate optimization
- Integrated AI/ML for predictive customer segmentation, automated anomaly detection, and personalization: built churn/retention model covering \$270M revenue with novel piecewise linear changepoint detection; LLM-based next-best-action agent generating personalized engagement recommendations; automated alerts on data refresh moving teams beyond descriptive reporting to prescriptive action
- Operate AI-first: built custom Codex tool stack (database access, SQL generation, schema introspection) for data science team, automating recurring analytical pipelines; taught Codex adoption across broader 20-person Data and AI group through executive sponsorship and regular peer training; drove company-wide Copilot adoption to 92% (~750 employees); personally matched output of 5-member team using agents

Director of Data Science

CareDx, Inc.

January 2024 – January 2025

San Francisco, CA

- Led multidisciplinary data organization (14 people, \$6M budget) spanning Bioinformatics, Statistics, and Data Science; managed managers; built multi-quarter roadmap with clear line-of-sight to business outcomes; set standards for modeling rigor, data quality, pipeline reliability, and production-readiness
- Directed production ML models (LSTM, SVM, LASSO) end-to-end from development through deployment, monitoring, and governance in regulated environment
- Oversaw enterprise analytics portfolio of ~20 Tableau dashboards serving Sales, Finance, Operations, and Clinical; established data governance framework for metric standards, instrumentation, and KPI integrity; built self-service analytics enabling cross-functional teams

Head of Product and Data Science

Cofactor Genomics (Healthcare / Diagnostics Startup)

December 2017 – January 2024

San Francisco, CA

- Owned end-to-end delivery of 2 production ML models: feature engineering, model optimization via bootstrap cross-validation, scalable cloud deployment with automated regression testing, CI/CD, and model monitoring
- Architected data platform from scratch: model repository (SageMaker), versioned training data (100s GBs), cloud pipelines (AWS Batch, +200% throughput, -25% cost), CI/CD (GitHub Actions), reconciliation pipelines ensuring data quality and accuracy at scale
- Designed and executed randomized controlled experimentation program across 12+ sites (~500 subjects): prespecified hypotheses, statistical power analysis, A/B testing methodology and cohort analysis informing pricing, positioning, and strategic decisions
- Led multidisciplinary team of 10 in high-growth startup; defined multi-quarter roadmaps and OKRs; partnered with Product, Design, Engineering, and Marketing on end-to-end user experience and data strategy
- First author on Nature ML paper (2nd most downloaded in 2022); authored 3 AI patents (2 granted)

CEO & Founder

May 2017 – August 2020

Where4Care (B2C Marketplace Startup)

St. Louis, MO

- Built two-sided marketplace product from zero: designed supplier-customer matching algorithm, full tech stack (Python/Django), cloud infrastructure; validated through 200-user study with behavioral instrumentation and conversion tracking

SELECTED PUBLICATIONS & PATENTS

3 Artificial Intelligence Patents (2 granted) – Methods for prediction, classification, and data processing using ML/AI
ML-based Prediction Model – *Nature Scientific Reports, 2022* (2nd most downloaded paper that year)

12 Peer-Reviewed Publications – Applied ML, deep learning, experimentation (*CVPR, ECCV, Nature, ICCP, 3DV*)

TECHNICAL SKILLS

Experimentation & Analytics: A/B/n Testing, Funnel Analysis, Conversion Rate Optimization, Causal Inference (DAGs, Causal Forests, DiD, Propensity Matching, Synthetic Controls), Behavioral Diagnostics, Cohort Analysis, Incrementality Analysis, Statistical Power Analysis

Predictive Analytics & AI/ML: Predictive Customer Segmentation, Automated Anomaly Detection, Personalization, Time Series Forecasting (ARIMA, Prophet), Churn/Retention Modeling, Propensity Scoring, Demand Forecasting

GenAI & Agentic Workflows: LLMs, RAG (Embeddings, Vector Search), Agentic Systems (Custom Tool Building, Codex), AI-Assisted Analytical Pipelines, AI-Native Team Operations

Data & Platform: Python, SQL, R, AWS (SageMaker, Batch), Databricks, ETL/Data Pipelines, Tableau, Streamlit, Data Governance (Metrics, Instrumentation, KPI Integrity), CI/CD, Docker, Git

Leadership: Strategic Ownership of Data Ecosystem, Building & Scaling High-Performing Teams (up to 14 reports, \$6M budget), Managing Managers, Executive Communication (C-Suite, Board), Cross-Functional Partnership (Product, Design, Engineering, Marketing, Finance, Executive)